Magnetic Moment in Magnetic Field

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Electric motors, speakers, and galvanometers operate based on the torque generated between magnetic moments and magnetic fields.

This experiment verifies

$$\vec{T} = \vec{m} \times \vec{B}$$

by measuring torque as a function of field strength, angle, and magnetic moment.

Introduction

From Lorentz Force to Torque:

1. Lorentz force on a moving charge:

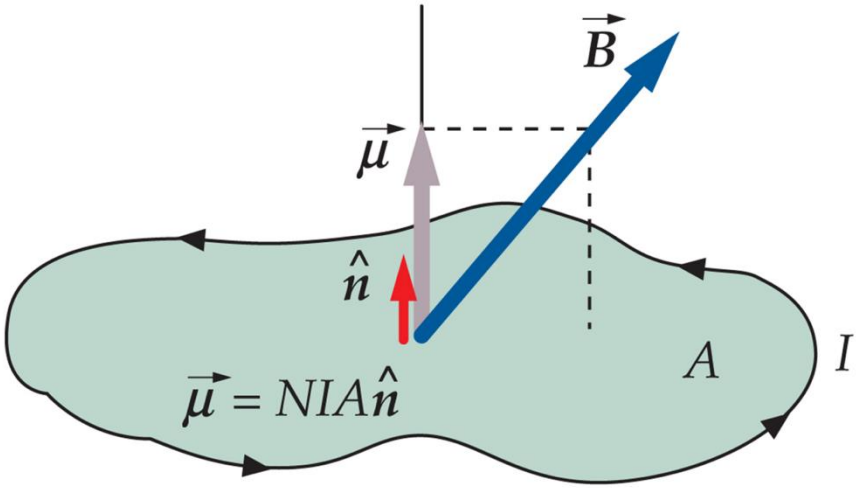
$$\vec{F} = q\vec{v} \times \vec{B}$$

2. For a current element $Id\vec{l}$:

$$d\vec{F} = Id\vec{l} \times \vec{B}$$

3. Torque about center:

$$d\vec{T} = \vec{r} \times d\vec{F} = \vec{r} \times (Id\vec{l} \times \vec{B})$$

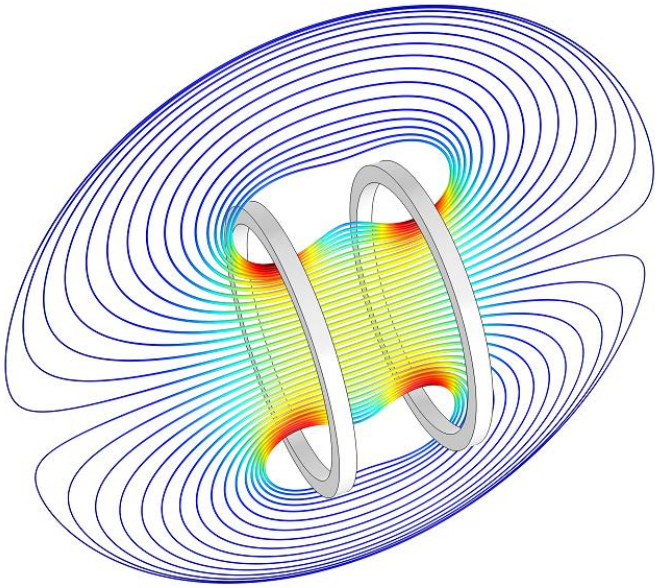


4. For a planar loop:

$$\vec{T} = I \oint \vec{r} \times (d\vec{l} \times \vec{B}) = IA\hat{n} \times \vec{B} = \vec{m} \times \vec{B}$$

where $\vec{m} = IA\hat{n}$ (direction by right-hand rule)

Helmholtz Configuration



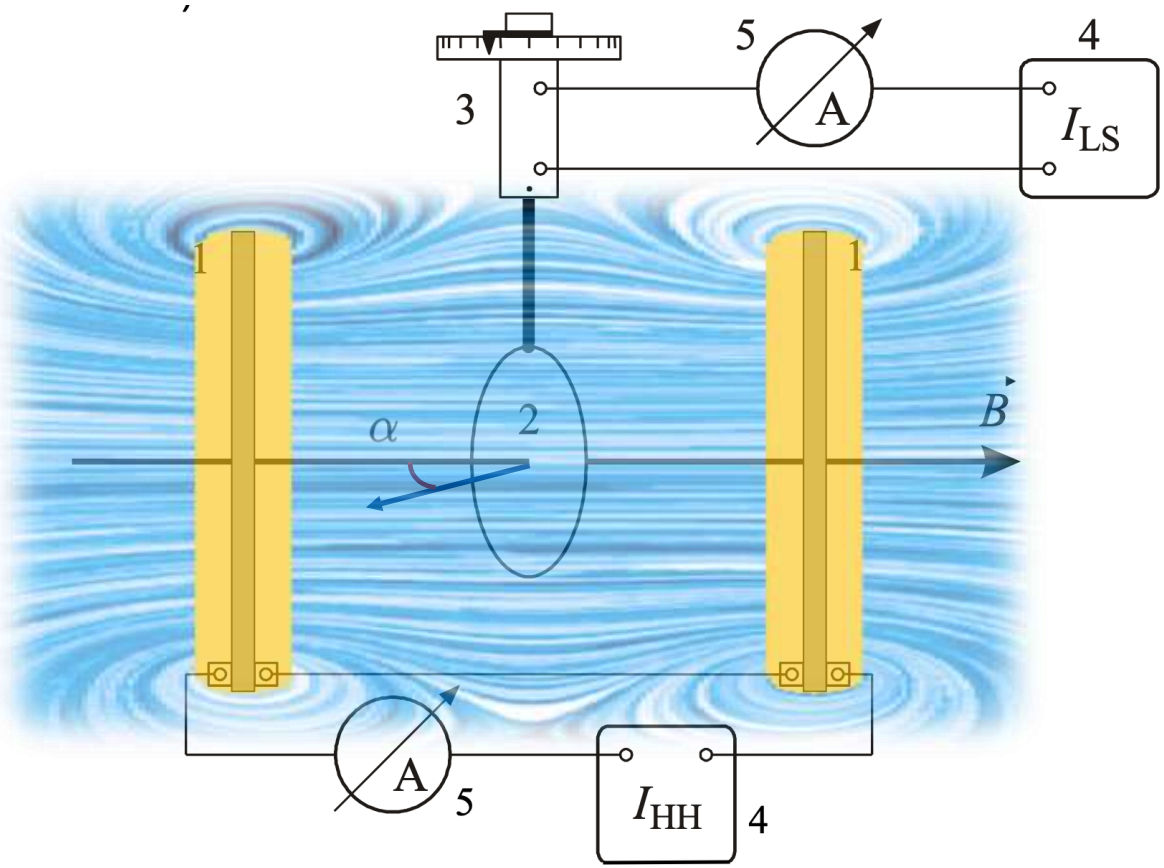
Two identical coils separated by their radius create a uniform field at the center:

$$B = \frac{8\mu_0 N I_H}{5\sqrt{5}R} = cI_H$$

For $N = 154$ turns and $R \approx 12.7$ cm:

$$c_{\text{theo}} \approx 6.8 \times 10^{-4}\ \text{T}\cdot\text{A}^{-1}$$

Experimental Setup

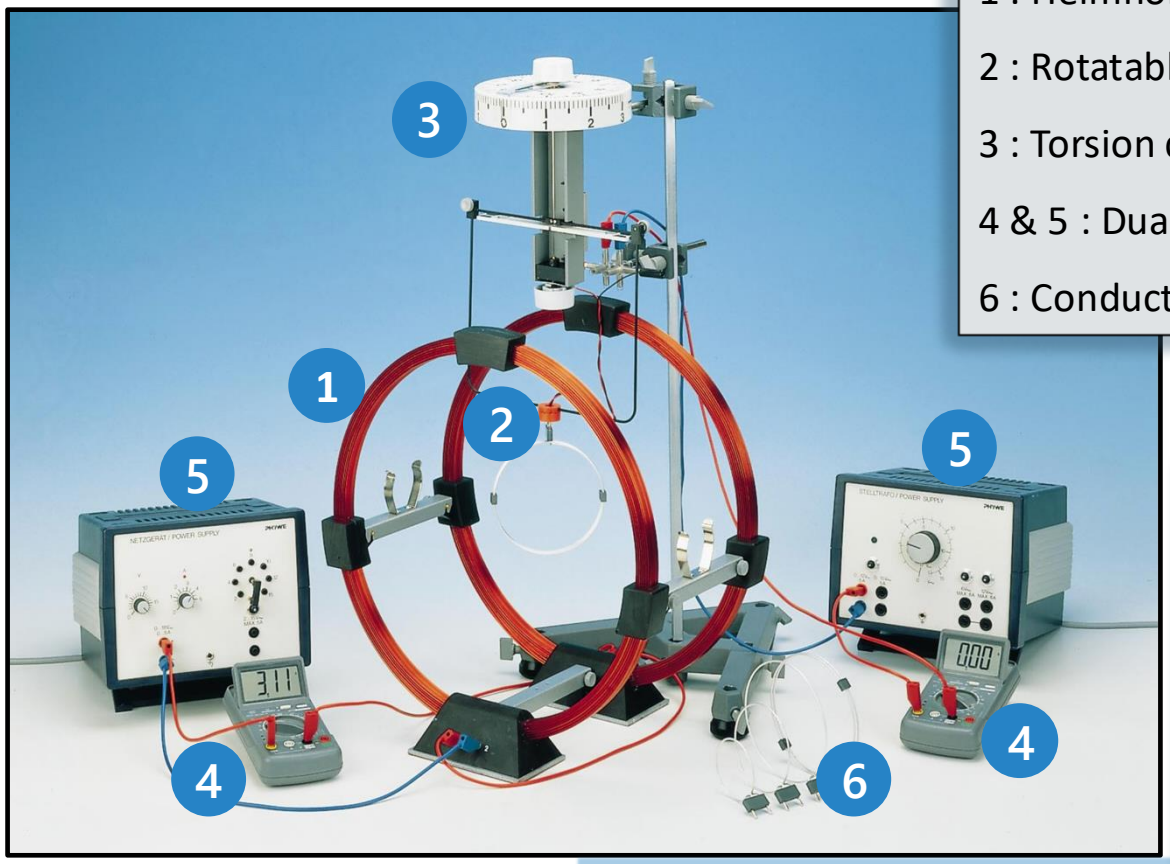


Schematic of Experimental Setup

$$\begin{cases} \vec{T} = \vec{m} \times \vec{B} \\ B = \dfrac{8\mu_0 N I_H}{5\sqrt{5}R} = cI_H \end{cases}$$

$$\Rightarrow \quad T = cI_H n I_L A \sin\alpha$$

Experimental Setup



Picture of Experimental Setup

Components:

1 : Helmholtz coils (N=154, R=20cm)

2 : Rotatable loop holder (15° increments)

3 : Torsion dynamometer (0.01 N resolution)

4 & 5 : Dual power supplies & multi-meters

6 : Conductor loop set (n = 1, 2, 3 turns)

Key specifications :

- Coil current: IH ≤ 3.0 A (continuous)
- Loop current: IL ≤ 3.0 A
- Angular resolution: 15°
- Force resolution: 0.01 mN

Experiment

$$T = cI_H n I_L A \sin\alpha \longrightarrow$$

Varied	Dependence
I_H	$T \propto I_H$ (linear)
n	$T \propto n$ (linear)
α	$T \propto \sin\alpha$ (sinusoidal)
d	$T \propto d^2$ (quadratic, since $A = \pi d^2/4$)
I_L	$T \propto I_L$ (linear)

Experimental Procedures:

1. Exp 1: Vary I_H (0-2.82 A), fixed loop (3 turns, 11.98 cm, 5 A, 90°)
2. Exp 2: Vary n (1,2,3 turns), fixed currents (2.5 A,5 A,90°)
3. Exp 3: Vary α (0°-90°), fixed config
4. Exp 4: Vary d (6.00, 8.47, 11.98 cm), fixed currents
5. Exp 5: Vary I_L (0-5 A), fixed field
6. Exp 6: Measure star loop (unknown area)

Raw Data

1. 3 loops

5.00A constant I_L

11.98 cm diameter (11.95, 12.00, 12.00 } averaged

$\alpha = 90° (\pm 1°)$

I_H/A	F_1/mV	F_2/mV	F_3/mV
0.00	0.00	0.00	0.00
0.50	0.70	0.60	0.65
1.00	1.10	1.10	1.10
1.50	1.65	1.60	1.60
2.00	2.20	2.20	2.15
2.50	2.70	2.70	2.70
2.82	3.05	3.10	3.05

error sources:

- placement of loop
- ⊥ of loop → drift wire too
- sagging of loop
  - ↳ zero value, measured value
- planar loops
- resistance by wire

2. 5.00A constant I_L

11.98cm diameter, same for all loops

$\alpha = 90° (\pm 1°)$

2.50A constant I_H

n	F_1/mV	F_2/mV	F_3/mV
1	0.90	0.90	0.90
2	1.80	1.80	1.80
3	2.70	2.65	2.70

3. 5.00A constant I_L

11.98 cm diameter

3 loops

2.50A constant I_H

$\alpha/°$	F_1/mV	F_2/mV	F_3/mV
0°	0.20	0.25	0.20
15°	1.00	0.95	0.90
30°	1.55	1.60	1.50
45°	2.05	2.00	2.15
60°	2.50	2.45	2.45
75°	2.70	2.70	2.65
90°	2.70	2.70	2.70

4. 5.00A constant I_L

2.50A constant I_H

$\alpha = 90° (\pm 1°)$

1 loop

d/cm	d_2/cm	d_3/cm	d avg/cm	F_1/mV	F_2/mV	F_3/mV
6.00	6.00	6.00	6.00	0.20	0.30	0.20
8.45	8.50	8.45	8.47	0.50	0.50	0.50
11.95	12.00	12.00	11.98	2.70	2.70	2.70

5. 2.50A constant I_H

$\alpha = 90° (\pm 1°)$

11.98 cm diameter

3 loops

I_L/A	F_1/mV	F_2/mV	F_3/mV
0.00	0.00	0.00	0.00
1.00	0.60	0.60	0.60
2.00	1.15	1.10	1.10
3.00	1.70	1.70	1.70
4.00	2.20	2.20	2.25
5.00	2.70	2.65	2.70

6. 5.00A constant I_L

2.50A constant I_H

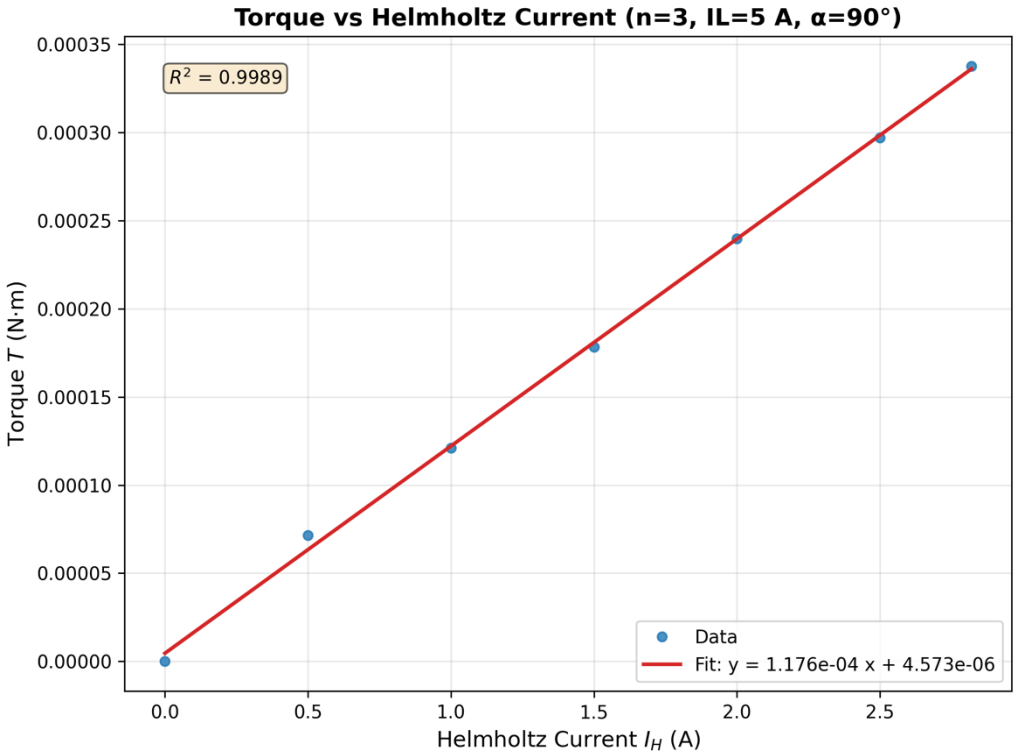
$\alpha = 90° (\pm 2°)$

~1 loop

A unknown

F_1/mV	0.85
F_2/mV	0.85
F_3/mV	0.85

Field Dependence



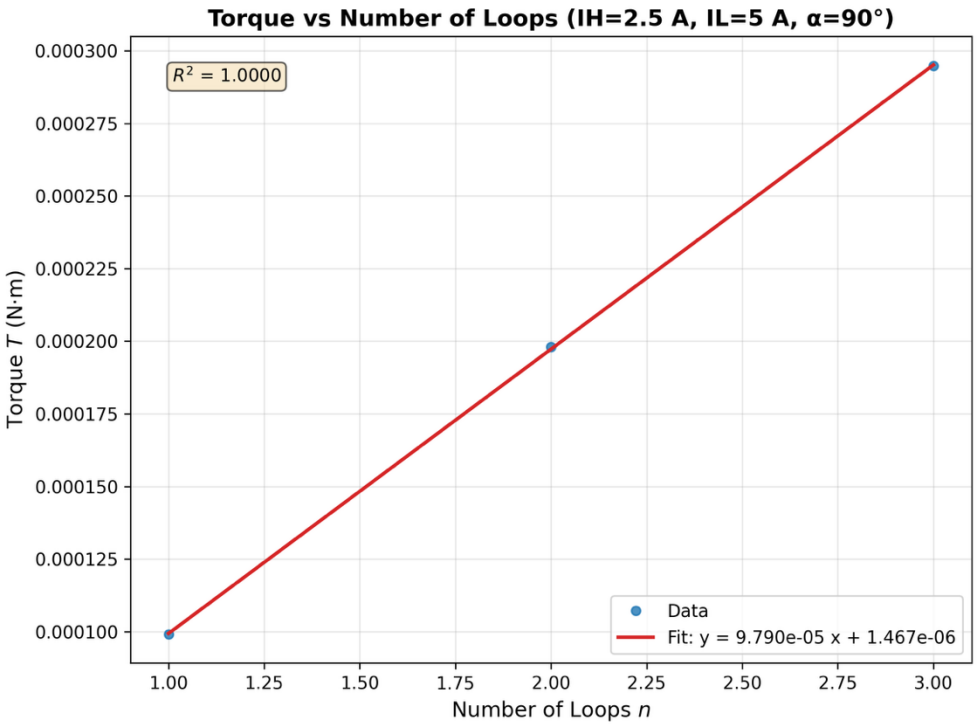
Configuration: 3-turn loop, $d = 11.98$ cm, $I_L = 5.00$ A, $\alpha = 90°$, I_H 0-2.82 A
Theory: $T = k_1 I_H$, with $k_1 = cnI_L A$
Results: $R^2 = 0.999$, slope $s_1 = 11.64 \times 10^{-5}$ N·m·A^{-1}
Helmholtz Constant Extraction:

$$A = \frac{\pi d^2}{4} = 1.127 \times 10^{-2} \text{ m}^2$$

$$c_1 = \frac{s_1}{nI_L A} = 6.95 \times 10^{-4} \text{ T·A}^{-1}$$

Interpretation: Torque increases linearly with magnetic field; no saturation up to 3 A.

Moment Scaling

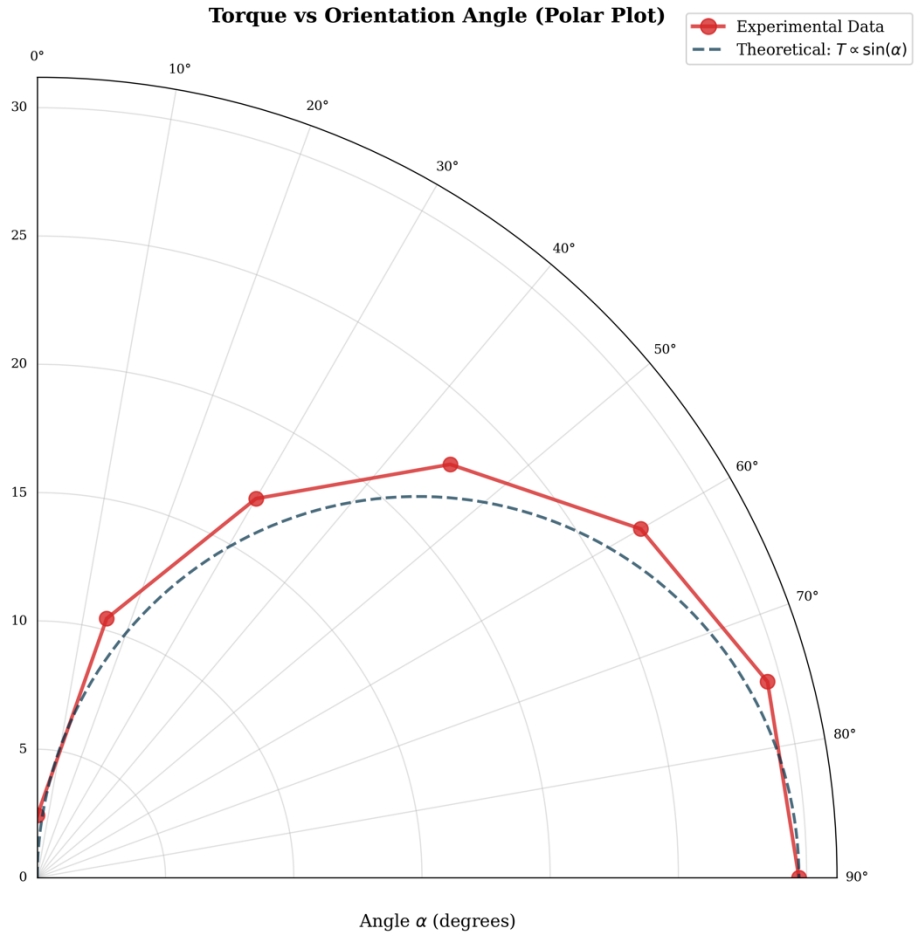


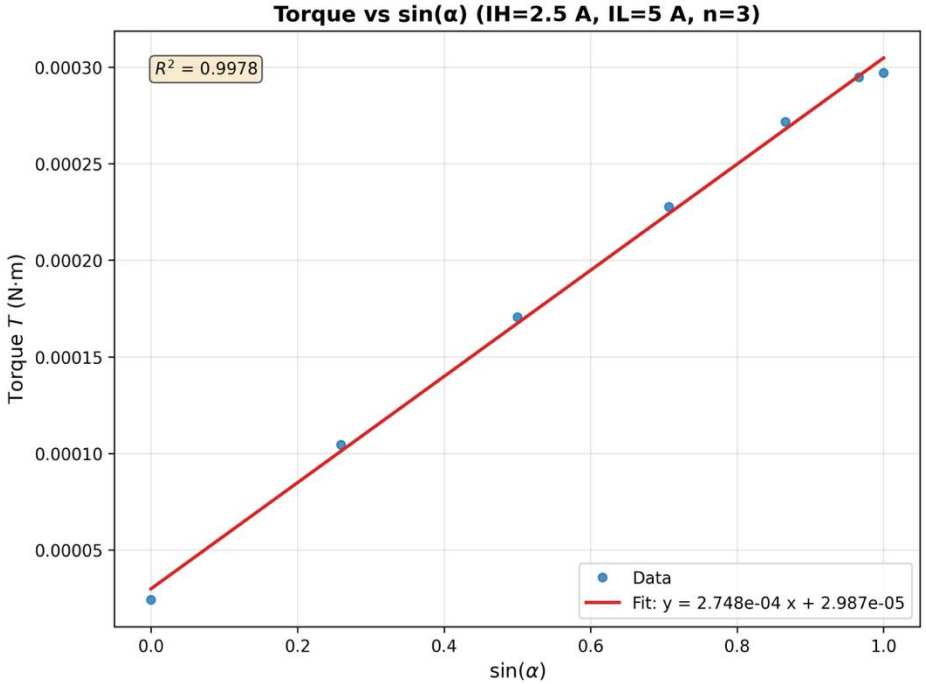
Configuration: $I_H = 2.5$ A, $I_L = 5.0$ A, $\alpha = 90°$, $n = 1, 2, 3$
Theory: $T = k_2 n$, linear dependence
Results: $R^2 = 1.000$, slope $s_2 = 9.87 \times 10^{-5}$ N·m

$$c_2 = \frac{s_2}{I_H I_L A} = 6.95 \times 10^{-4}\ \text{T·A}^{-1}$$

Interpretation: Perfect linearity; each turn contributes equally to total moment.

Angular Dependence





Configuration: 3-turn loop, $d = 11.98$ cm, $I_H = 2.5$ A, $I_L = 5$ A, $\alpha = 0° - 90°$

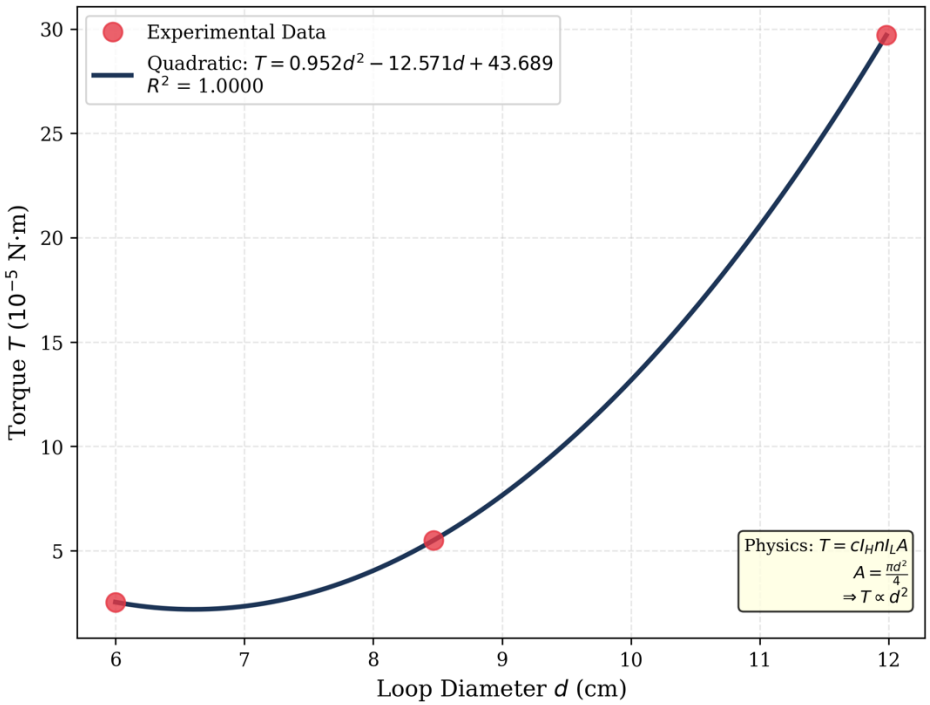
Theory: $T = k_3 \sin\alpha$, $k_3 = cI_H n I_L A$

Results: $R^2 = 0.998$, slope $s_3 = 29.12 \times 10^{-5}$ N·m

$$c_3 = \frac{s_3}{I_H n I_L A} = 6.50 \times 10^{-4}\ \text{T·A}^{-1}$$

Maximum torque at 90°, small reading at 0° due to alignment error (8%).

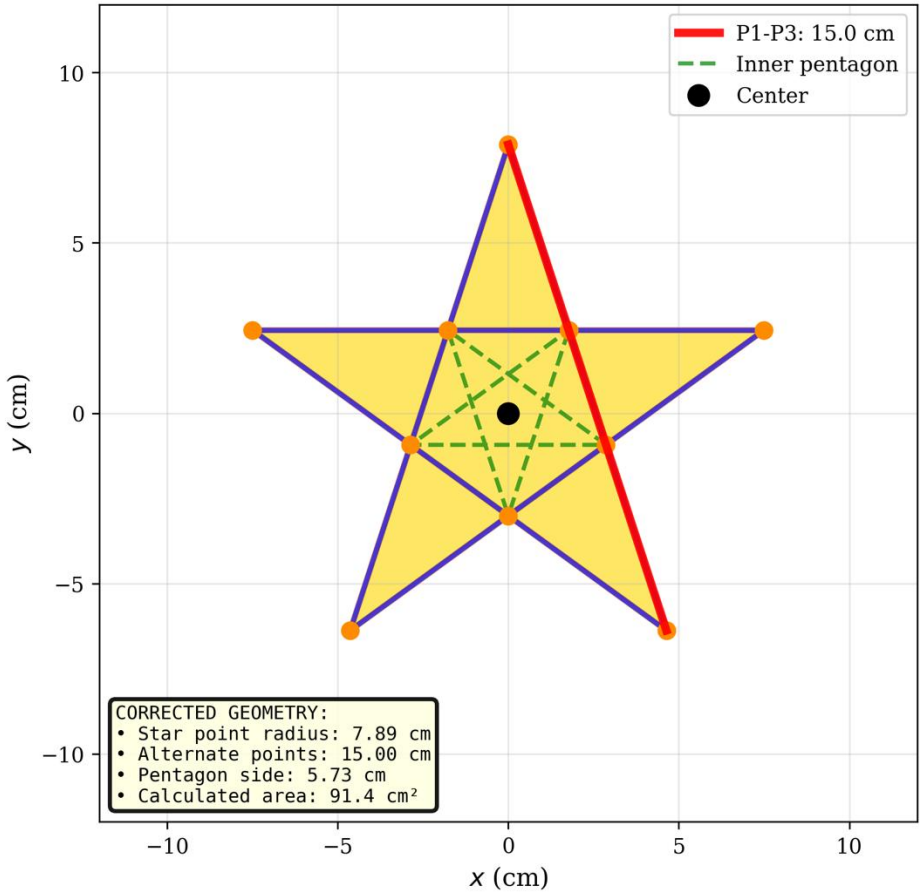
Diameter Dependence



Configuration: single-turn loops, $d = 6.00, 8.47, 11.98$ cm, $I_H = 2.5$ A, $I_L = 5$ A, $\alpha = 90°$
Theory: $T = cI_H n I_L(\pi d^2/4)\sin\alpha$, quadratic dependence
Results: Quadratic fit $R^2 = 0.997$, coefficient $a = 0.208$ N·m·cm^{-2}

$$c_4 = \frac{a \times 10^4}{I_H n I_L(\pi/4)} = 6.63 \times 10^{-4}\ \text{T·A}^{-1}$$

Star Loop Analysis

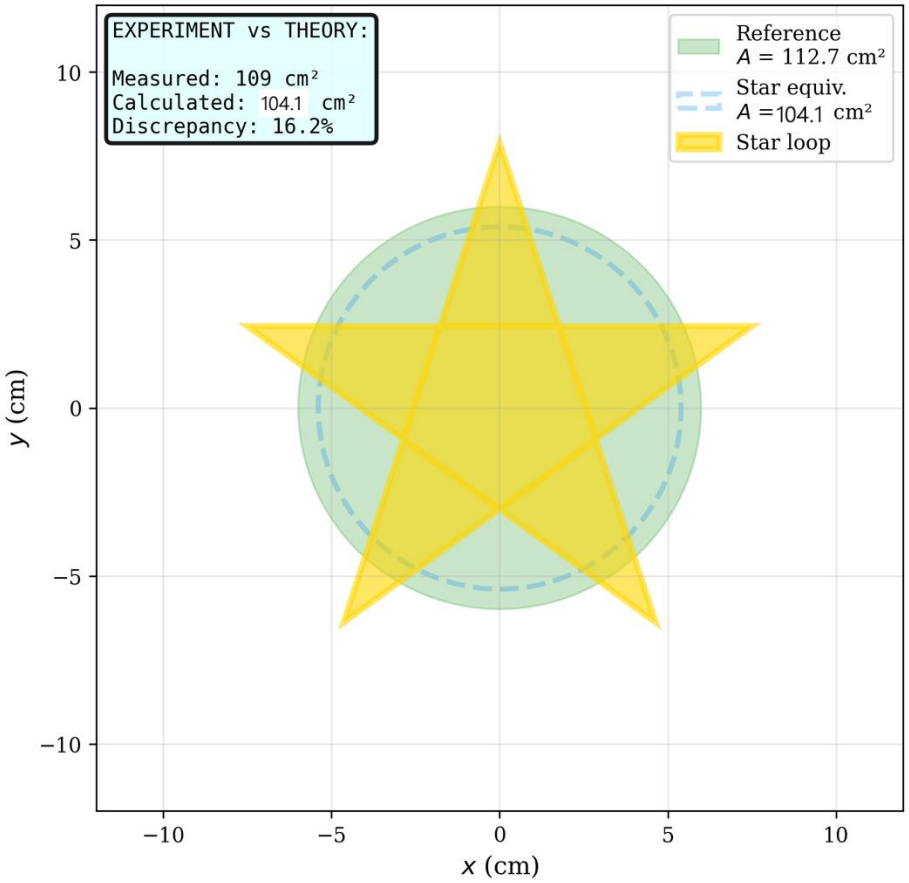


Stokes' theorem: for planar loop, magnetic moment depends only on enclosed area, independent of shape:

$$m = IA \quad \text{(shape-independent)}$$

A star and circle with same area → same magnetic moment → same torque in uniform field.

Star Loop Analysis



Configuration: $I_H = 2.5$ A, $I_L = 5$ A, $n \approx 1$, $\alpha = 90°$, measured $T = (9.35 \pm 0.11) \times 10^{-5}$ N · m

Area Determination:

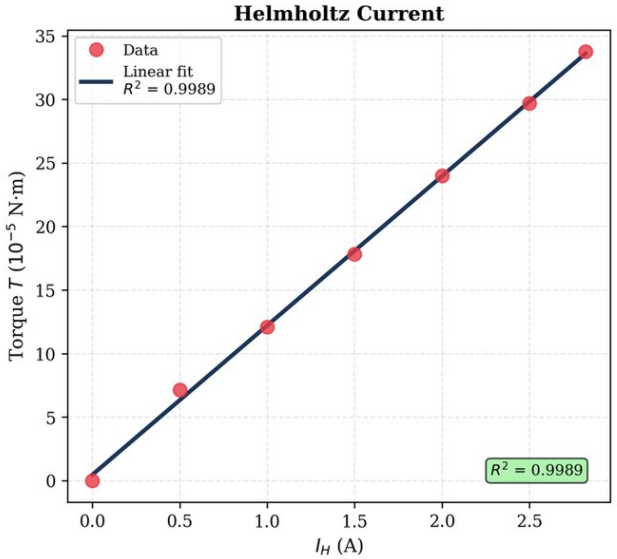
$$A_{\text{star}} = \frac{T}{cI_H n I_L \sin\alpha} = 1.10 \times 10^{-2}\ \text{m}^2 = 110\ \text{cm}^2$$

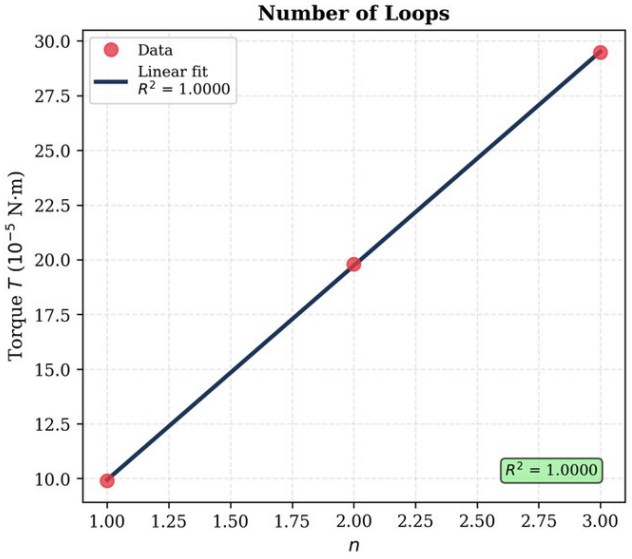
Equivalent Circular Diameter:

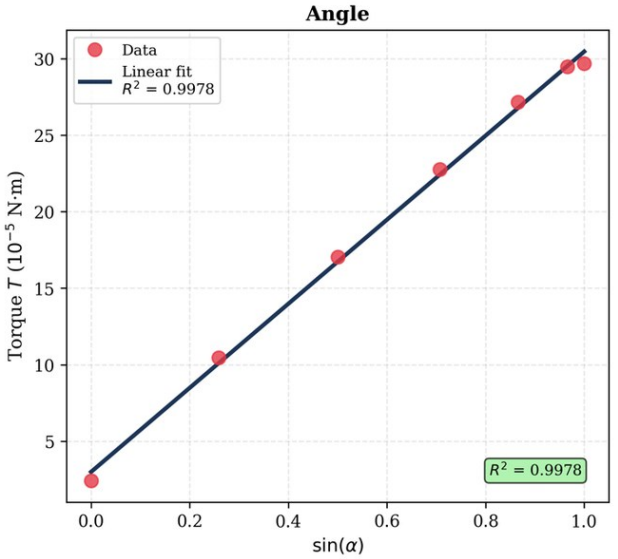
$$d_{\text{equiv}} = \sqrt{4A/\pi} = 11.8\ \text{cm}$$

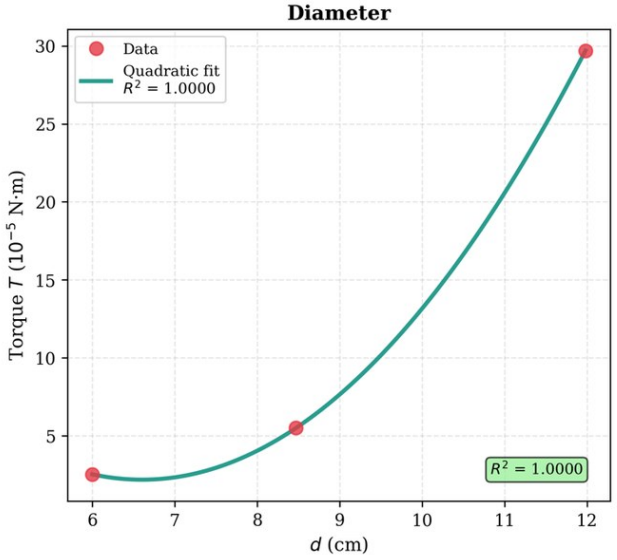
Geometric Validation: Star loop measured geometrically = 104.1 cm^2; experimental effective area higher due to possible turn count, out-of-plane bending, or fabrication tolerances.

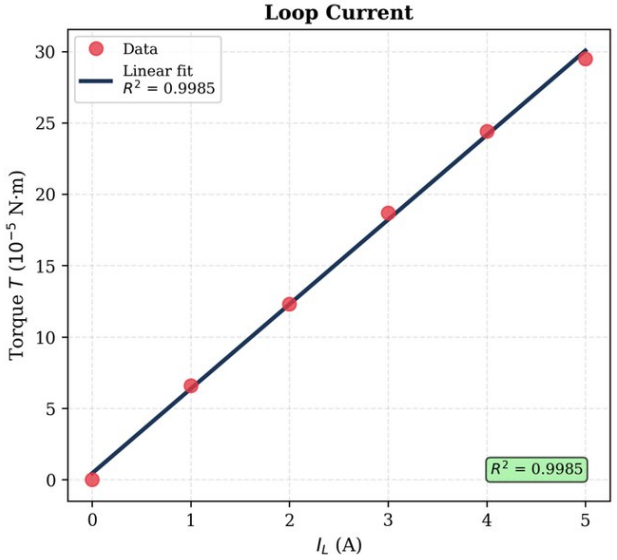
Discussion












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SCALING LAW VALIDATION

Expected Relationships:
• T ∝ IH (linear)
• T ∝ n (linear)
• T ∝ sin(α) (linear)
• T ∝ d² (quadratic)
• T ∝ IL (linear)

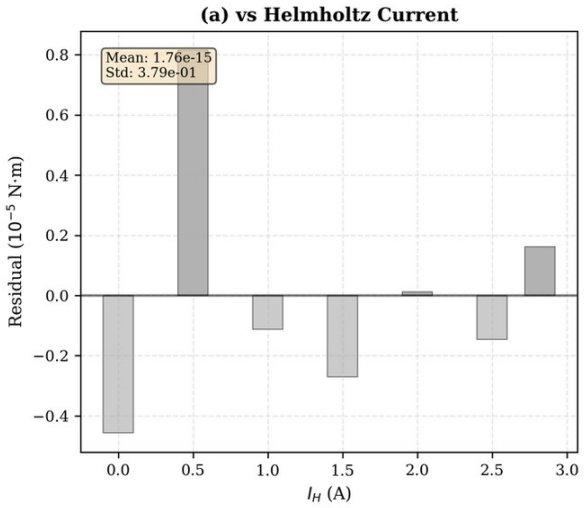
Master Equation:
T = c·IH·n·IL·A·sin(α)

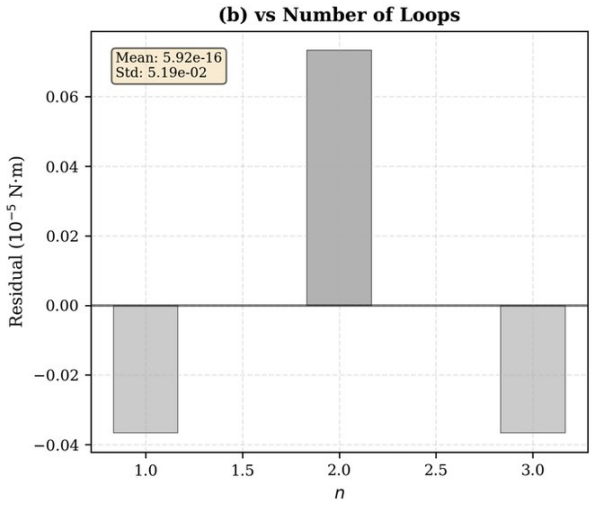
where A = πd²/4 for circles

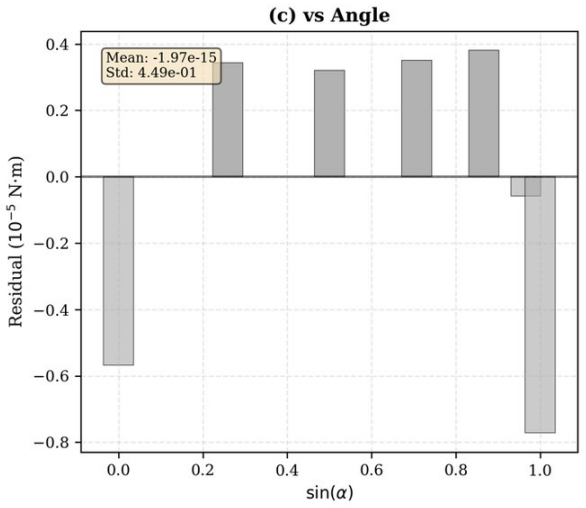
Color Legend:
□ R² > 0.995: Excellent
□ R² > 0.950: Good
□ R² < 0.950: Poor

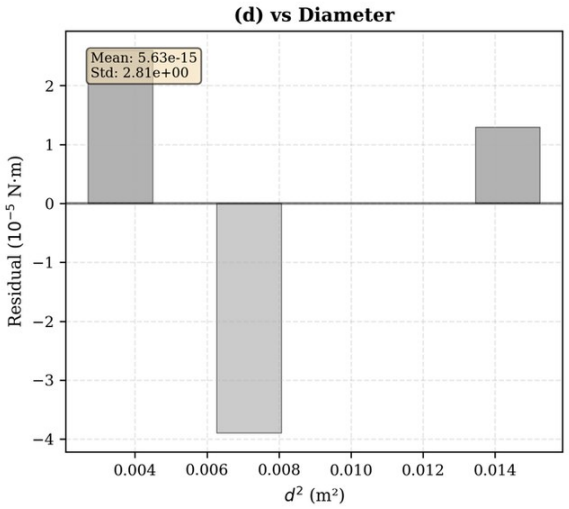
All relationships validated
within experimental precision.
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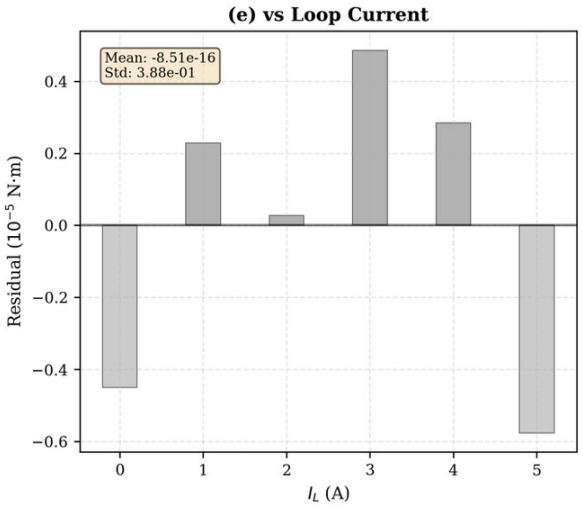
Residual Analysis: Deviations from Linear Fits











Residuals scatter randomly around zero with no trends. Magnitudes (≈ 10^{-6} N·m) match instrument resolution.

Sources of Error

- **Torsion Balance Instability:** The dynamometer is highly sensitive and prone to zero-point drift. The connecting wires also introduce an unquantifiable parasitic torque that biases the $\vec{T}$ measurement.
- **Non-Uniform Magnetic Field ($\vec{B}$):** The ideal Helmholtz arrangement is compromised by any coil misalignment or incorrect separation (R). This field non-uniformity across the loop violates the core assumption of the theoretical torque equation.
- **Non-Ideal Loop Geometry (Planarity):** The star-shaped loop is often not perfectly planar, resulting in an ill-defined magnetic area vector ($\vec{A}$). Furthermore, systematic errors in measuring dimensions (d and R) propagate into constant calculations.
- **Inaccurate Angle Setting (α):** Measurements rely on coarse 15° physical intervals. This low, discrete resolution in setting the angle contributes to significant scatter in the $\vec{T}$ vs. $\sin\alpha$ data.

Conclusion

- The experiment **validated the theoretical model** for magnetic torque on current loops in a uniform magnetic field.
- Five independent parameter sweeps confirmed the linear dependencies predicted by Equation 5:

$$T = cI_H n I_L A \sin\alpha$$

with correlation coefficients $R^2 > 0.997$ for all experiments using the correct functional forms.

- The Helmholtz field constant was calibrated through cross-validated measurements, yielding:

$$c = (6.81 \pm 0.23) \times 10^{-4}\,\mathrm{T \cdot A^{-1}} \quad (3.3\%\ \text{uncertainty})$$

- Using the calibrated field constant, the effective magnetic area of an unknown star-shaped loop was determined as:

$$A_{\text{eff}} = (110 \pm 4)\,\mathrm{cm^2}$$

This is in close agreement with the geometrically calculated area of $104.1\,\mathrm{cm^2}$, demonstrating that magnetic moment depends only on the enclosed area, **independent of loop shape** (consistent with Stokes' theorem).

Thank You!

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Open for Questions